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- **Term:** Term 3, AY24-25

Required Python Packages

Run the following applications once if they are not yet installed in your Python environment:

In [1]:
```
!pip install numpy
```

```
Requirement already satisfied: numpy in c:\users\naval\anaconda3\lib\site-packages
(2.3.5)
```

In [2]:
```
!pip install scipy
```

```
Requirement already satisfied: scipy in c:\users\naval\anaconda3\lib\site-packages
(1.16.3)
Requirement already satisfied: numpy<2.6,>=1.25.2 in c:\users\naval\anaconda3\lib\si
te-packages (from scipy) (2.3.5)
```

In [3]:
```
!pip install matplotlib
```

```
Requirement already satisfied: matplotlib in c:\users\naval\anaconda3\lib\site-packa
ges (3.10.6)
Requirement already satisfied: contourpy>=1.0.1 in c:\users\naval\anaconda3\lib\site
-packages (from matplotlib) (1.3.3)
Requirement already satisfied: cycler>=0.10 in c:\users\naval\anaconda3\lib\site-pac
kages (from matplotlib) (0.11.0)
Requirement already satisfied: fonttools>=4.22.0 in c:\users\naval\anaconda3\lib\sit
e-packages (from matplotlib) (4.60.1)
Requirement already satisfied: kiwisolver>=1.3.1 in c:\users\naval\anaconda3\lib\sit
e-packages (from matplotlib) (1.4.9)
Requirement already satisfied: numpy>=1.23 in c:\users\naval\anaconda3\lib\site-pack
ages (from matplotlib) (2.3.5)
Requirement already satisfied: packaging>=20.0 in c:\users\naval\anaconda3\lib\site-
packages (from matplotlib) (25.0)
Requirement already satisfied: pillow>=8 in c:\users\naval\anaconda3\lib\site-packag
es (from matplotlib) (12.0.0)
Requirement already satisfied: pyparsing>=2.3.1 in c:\users\naval\anaconda3\lib\site
-packages (from matplotlib) (3.2.5)
Requirement already satisfied: python-dateutil>=2.7 in c:\users\naval\anaconda3\lib
\site-packages (from matplotlib) (2.9.0.post0)
Requirement already satisfied: six>=1.5 in c:\users\naval\anaconda3\lib\site-package
s (from python-dateutil>=2.7->matplotlib) (1.17.0)
```

In [4]:
```
!pip install panda
```

```
Requirement already satisfied: panda in c:\users\naval\anaconda3\lib\site-packages
(0.3.1)
Requirement already satisfied: setuptools in c:\users\naval\anaconda3\lib\site-packa
ges (from panda) (80.9.0)
Requirement already satisfied: requests in c:\users\naval\anaconda3\lib\site-package
s (from panda) (2.32.5)
Requirement already satisfied: charset_normalizer<4,>=2 in c:\users\naval\anaconda3
\lib\site-packages (from requests->panda) (3.4.4)
Requirement already satisfied: idna<4,>=2.5 in c:\users\naval\anaconda3\lib\site-pac
kages (from requests->panda) (3.11)
Requirement already satisfied: urllib3<3,>=1.21.1 in c:\users\naval\anaconda3\lib\si
te-packages (from requests->panda) (2.6.3)
Requirement already satisfied: certifi>=2017.4.17 in c:\users\naval\anaconda3\lib\si
te-packages (from requests->panda) (2026.1.4)
```

In [5]:
```
!pip install gekko
```

```
Requirement already satisfied: gekko in c:\users\naval\anaconda3\lib\site-packages
(1.3.2)
Requirement already satisfied: numpy>=1.8 in c:\users\naval\anaconda3\lib\site-packa
ges (from gekko) (2.3.5)
```

In [6]:
```
!pip install scikit-learn
```

```
Requirement already satisfied: scikit-learn in c:\users\naval\anaconda3\lib\site-pac
kages (1.7.2)
Requirement already satisfied: numpy>=1.22.0 in c:\users\naval\anaconda3\lib\site-pa
ckages (from scikit-learn) (2.3.5)
Requirement already satisfied: scipy>=1.8.0 in c:\users\naval\anaconda3\lib\site-pac
kages (from scikit-learn) (1.16.3)
Requirement already satisfied: joblib>=1.2.0 in c:\users\naval\anaconda3\lib\site-pa
ckages (from scikit-learn) (1.5.2)
Requirement already satisfied: threadpoolctl>=3.1.0 in c:\users\naval\anaconda3\lib
\site-packages (from scikit-learn) (3.5.0)
```

In [7]:
```
!pip install sympy
```

```
Requirement already satisfied: sympy in c:\users\naval\anaconda3\lib\site-packages
(1.14.0)
Requirement already satisfied: mpmath<1.4,>=1.1.0 in c:\users\naval\anaconda3\lib\si
te-packages (from sympy) (1.3.0)
```

Machine Problem 001: Laplace Transforms and Sympy

Laplace transforms convert a function $f(t)$ in the time domain into function in the Laplace domain $F(s)$.

$$F(s) = \mathcal{L}\left(f(t)\right) = \int_0^\infty f(t)e^{-s\,t}dt$$

The SymPy library is a symbolic mathematics library written for Python that can be used to perform analytical calculations. This library can be very useful for control system design and analysis.

Table of Common Laplace Transforms

Mathematicians have developed tables of commonly used Laplace transforms. Here is summary table with a few of the entries that will be most common for analysis of linear differential equations.

Item no.	f(t)	F(s)
1.	$\delta(t)$	1
2.	$u(t)$	$\frac{1}{s}$
3.	$tu(t)$	$\frac{1}{s^2}$
4.	$t^n u(t)$	$\frac{n!}{s^{n+1}}$
5.	$e^{-at}u(t)$	$\frac{1}{s+a}$
6.	$\sin(\omega t)u(t)$	$\frac{\omega}{s^2+\omega^2}$
7.	$\cos(\omega t)u(t)$	$\frac{s}{s^2+\omega^2}$

In [8]:
```
import sympy as sym
from sympy.abc import s,t,u,x,y,z,a,b,c,k
from sympy.integrals import laplace_transform
```

In [9]:
```
# For pretty  printing
sym.init_printing()
```

In [10]:
```
# Define variables for time 't' and complex variable 's'
# t = sym.symbols('t', real=True)
# s = sym.symbols('s')

# Define a constant:
# a = sym.symbols('a', real=True, positive=True)
```

1. Evaluate:

$$\mathcal{L}\{\delta(t)\} = 1$$

As an example of the Laplace transform, consider a constant 1. The function $f(t) = 1$ and the following expression is integrated.

$$\mathcal{L}(1) = \int_0^\infty 1\, e^{-s\,t} dt = -\frac{1}{s} e^{-s\,t}\Big|_0^\infty = 0 - \left(-\frac{1}{s}\right) = \frac{1}{s}$$

Notice that the derived value for a constant 1 is the unit step function with $f(t) = 1$ where a signal output changes from 0 to 1 at $time = 0$.

In [11]:
```
# Define the time domain function:
f1 = 1
f1
```

Out[11]: 1

In [12]:
```
# Compute the Laplace Transform:
f1s = laplace_transform(f1, t, s, noconds=True)
f1s
```

Out[12]: $\frac{1}{s}$

2. Evaluate:

$$\mathcal{L}\{u(t)\} = \frac{1}{s}$$

$$\mathcal{L}\{t\} = \frac{1}{s^2}$$

In [13]:
```
# Define the time domain function:
f2 = t
f2
```

Out[13]: t

In [14]:
```
# Compute the Laplace Transform:
f2s = laplace_transform(f2, t, s, noconds=True)
f2s
```

Out[14]: $\frac{1}{s^2}$

3. Evaluate:

$$\mathcal{L}\{tu(t)\} = \frac{1}{s^2}$$

$$\mathcal{L}\{5t\} = \frac{5}{s^2}$$

In [15]:
```
# Laplace transform (t->s)
U = laplace_transform(5*t, t, s)
```

```
U[0]
```

Out[15]: $\frac{5}{s^2}$

4. Evaluate:

$$\mathcal{L}\{t^n u(t)\} = \frac{n!}{s^{n+1}}$$

$$\mathcal{L}\{t^2\} = \frac{2}{s^3}$$

In [16]:
```
# Define the time domain function:
f4 = t**2
f4
```

Out[16]: t^2

In [17]:
```
# Compute the Laplace Transform:
f4s = laplace_transform(f4, t, s, noconds=True)
f4s
```

Out[17]: $\frac{2}{s^3}$

5. Evaluate:

$$\mathcal{L}\{e^{-at}u(t)\} = \frac{1}{s+a}$$

Solution:

$$\mathcal{L}\{e^{-at}\} = \int_0^\infty e^{-a\,t} e^{-s\,t} dt$$

$$= \int_0^\infty e^{-(s+a)\,t} dt$$

$$= \frac{e^{-(s+a)t}}{-(s+a)}\Bigg|_0^\infty$$

$$= 0 - \left(-\frac{1}{s+a}\right) = \frac{1}{s+a}$$

In [18]:
```
# Define the time domain function:
f5 = sym.exp(-a*t)
f5
```

Out[18]: e^{-at}

In [19]:
```
# Compute the Laplace Transform:
f5s = laplace_transform(f5, t, s, noconds=True)
f5s
```

Out[19]: $\frac{1}{a+s}$

6. Evaluate:

$$\mathcal{L}\{\sin(\omega t)u(t)\} = \frac{\omega}{s^2+\omega^2}$$

$$\mathcal{L}\{\sin(4t)\} = \frac{4}{s^2+16}$$

In [20]:
```
# Define the time domain function:
f6 = sym.sin(4*t)
f6
```

Out[20]: $\sin(4t)$

In [21]:
```
# Compute the Laplace Transform:
f6s = laplace_transform(f6, t, s, noconds=True)
f6s
```

Out[21]: $\frac{4}{s^2+16}$

7. Evaluate:

$$\mathcal{L}\{\cos(\omega t)u(t)\} = \frac{s}{s^2+\omega^2}$$

$$\mathcal{L}\{\cos(\frac{t}{2})\} = \frac{s}{s^2+\frac{1}{4}}$$

In [22]:
```
# Define the time domain function:
f7 = sym.cos(t/2)
f7
```

Out[22]: $\cos\left(\frac{t}{2}\right)$

In [23]:
```
# Compute the Laplace Transform:
f7s = laplace_transform(f7, t, s, noconds=True)
f7s
```

Out[23]: $\frac{s}{s^2+\frac{1}{4}}$

In [23]:

```

```